ABREHAM DESSU

Full Stack Software Engineer

abreham.dessu@gmail.com +251 931 521 167 linkedin.com/in/abrehamdessu github.com/abriz19

SUMMARY

Full Stack Engineer with 4+ years in TypeScript — React and Next.js on the client, Node.js and NestJS APIs over PostgreSQL underneath. Synchronized 5,000+ CRM records with near-zero drift, built payment infrastructure clearing 5,000+ transactions a day at a 96% success rate, grew a Kafka microservices platform to 10,000+ users, and cut downtime 40%. Owns features from schema design through deployment, and works AI tooling into daily delivery without loosening review standards.

EXPERIENCE

Software Engineer
06/2025 – Present

ETM Software
Addis Ababa, Ethiopia

Own features end to end across the client, server, and database layers — TypeScript on NestJS and PostgreSQL, deployed on Google Cloud Platform.

- Improved system reliability 30% by re-architecting the backend service layer and replacing reactive firefighting with production monitoring, alerting, and automated CI/CD pipelines on GCP.
- Kept 5,000+ HubSpot CRM records in sync with near-zero drift by building a fault-tolerant integration with retry, deduplication, and a periodic reconciliation pass that survives API rate limits and partial failures.
- Eliminated manual data entry across 2,000+ hierarchical entities by designing the PostgreSQL data models and automating multi-level synchronisation, removing a recurring source of data corruption.
- Cut feature delivery time 25% by designing 5+ reusable service modules in NestJS and Prisma ORM, establishing shared data-access and validation patterns other engineers now build on.
- Shipped GraphQL and REST APIs that collapsed multi-call data fetching into single round trips, with Jest and Supertest coverage on every endpoint.
- Zero authentication-related incidents to date under an access model built on Firebase Authentication, JWT, and role-based permissions.

Full Stack Developer
09/2023 – 06/2025

Ablaze Labs
Addis Ababa, Ethiopia

Built product across four platforms — a gaming and payments platform, a national government CMS, a job board, and a medical e-learning system — on both the React frontends and the Node.js services behind them.

- Scaled payment infrastructure to 5,000+ daily transactions at a 96% success rate, integrating Chapa and SantimPay with idempotent request handling, webhook signature verification, and automated failure recovery.
- Cut production downtime 40% and improved application response times 20% by profiling slow queries, tuning PostgreSQL indexes, and resolving connection-pool saturation that was starving unrelated requests.
- Grew an event-driven platform to 10,000+ users, designing Kafka-based microservices that decoupled payments, referrals, and user management so each could scale and fail independently.
- Led delivery of a national CMS for Ethiopia's Ministry of Water and Energy across 12 interconnected subprojects, owning both the shared backend API layer and the frontend content management modules.
- Replaced manual, document-driven government reporting with automated APIs and CMS interfaces, cutting report turnaround from 3 days to under 2 hours.
- Shipped a job board end to end — employer and applicant workflows, subscription billing, and a referral engine — on server-rendered Next.js backed by NestJS services.

Junior Software Developer
10/2022 – 08/2023

Hirundo Security Solution
Addis Ababa, Ethiopia

First engineering role: internal tools and automation for an IT and security services business.

- Built an inventory management system to track the company's security hardware — CCTV cameras, fingerprint scanners, and biometric access devices — covering stock levels, assignment to client sites, and maintenance history.

- Improved deployment efficiency 22% by scripting configuration and provisioning across 10+ machines, replacing a manual multi-step setup with a repeatable workflow.
- Raised IT support efficiency 15% with internal web tools that automated repetitive manual processes, and contributed unit and integration tests and bug fixes to existing systems.

EDUCATION

Software Engineering Trainee — Data Structures and Algorithms 01/2025 – 01/2026
A2SV (Africa to Silicon Valley) Addis Ababa, Ethiopia

- Solved 300+ algorithmic problems in Python across LeetCode, Codeforces, and HackerRank — arrays, strings, linked lists, stacks, queues, trees, graphs, recursion, and dynamic programming.
- Focused on time and space complexity analysis, systematically optimising solutions from brute-force to efficient approaches.

B.Sc. Electrical and Computer Engineering 11/2016 – 07/2022
Adigrat University Adigrat, Ethiopia

SKILLS

Primary Stack
TypeScript • Node.js • NestJS • React • Next.js • PostgreSQL

Also Work With
JavaScript • Python • Go • SQL • Redis • MongoDB • Prisma ORM • Angular • Tailwind CSS

Reliability & Testing
Prometheus • Grafana • Production Monitoring • Incident Response • Jest • Vitest • Supertest

AI Tooling
Claude Code • Cursor • GitHub Copilot — used daily for code generation, testing, and review

Architecture & APIs
REST API Design • GraphQL • Microservices • Event-Driven Architecture • Apache Kafka • Asynchronous Processing • Third-Party Integrations • Webhooks

Cloud & Infrastructure
Google Cloud (Cloud Run, Cloud Build) • AWS • Docker • Kubernetes • Terraform • GitHub Actions • CI/CD

Security
OAuth 2.0 • JWT • Role-Based Access Control • Firebase Authentication

Soft Skills
Remote & Asynchronous Collaboration • Cross-Functional Teamwork • Stakeholder Communication • Requirements Translation • Technical Documentation • Code Review & Constructive Feedback • Autonomous Ownership

CERTIFICATIONS

AWS Certified Solutions Architect – Associate
Amazon Web Services

Microsoft Azure Fundamentals (AZ-900)
Microsoft, Aug 2025

AWS Scholars: Build Your First AI Productivity App
Udacity, Mar 2026

AI Career Essentials
ALX Africa, Jul 2024

AWS Certified Cloud Practitioner
Amazon Web Services, Aug 2025

AWS Scholars: Analyze Data using AI with PartyRock
Udacity, Mar 2026

Professional Foundations
ALX Africa, Apr 2025

IELTS Academic — Speaking Band 7.0
British Council, Oct 2025